What Has the Impact of the Epidemic Brought to Related Party Companies?

Visual Analysis Based on the Perspective of “Space-Industry-Time” of A-Share Data in 2017~2024:

A Case Study of Companies in Guangdong Province

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1 Introduction

This study takes A-share listed companies in Guangdong Province from 2017 to 2024 as a sample to analyze the scale, structure, and evolution pattern of related party transactions (RPTs) around the systemic impact caused by COVID-19 [1]. As an important part of the capital market within an enterprise group, RPT may play the role of "risk buffer and stabilizer" in crisis situations and may also be used for "resource transfer and tunneling behavior".

This study constructs a three-dimensional visualization framework of space (regional)-industry-temporal, and compares the dynamic changes of RPT before the epidemic (2017–2019), during the epidemic (2020–2022), and in the post-epidemic period (2023–2024), in order to reveal the internal trading strategies [2], governance responses [3], and network dependence characteristics of enterprises under exogenous shocks [4], and combine financial data and network analysis [5] to provide a basis for subsequent causal testing.

1.1 Concept and Theoretical Framework of Related Party Transactions

RPT is an important channel for resource reallocation within the group [6]. When external financing tightens or transaction costs rise, it may play a "buffer" function or create a tunneling effect. COVID-19, as a typical exogenous shock, has led to supply chain disruptions and increased uncertainty, further catalyzing the scale expansion and structural changes of RPT [9]. Research and regulatory evidence show that intra-business transactions have shown two significant trends during the pandemic:

(1) Support mechanism: risk sharing and operational stability. Stabilize operations and alleviate financial pressure through internal loans, guarantees, and internal procurement [7], and play a buffer function. This results in the concentration of key operations within the group to reduce supply chain risks. Internal capital adjustment enhances the ability to support subsidiaries, and large groups strengthen capital exchanges with controlling shareholders to hedge against impacts.

(2) Tunnel behavior: resource transfer and information concealment. In the context of weak regulation, it has become a tool for controlling shareholders to convey interests and occupy funds, which produces a tunnel effect [8]. As a result, the number of violated RPTs and penalties has increased, and covert methods such as factoring and fictitious agreements have increased, and the epidemic has led to an increase in information asymmetry, making it more difficult to identify unfair transactions.

1.2 Research Scope

The core hypothesis is that the scale and frequency of RPT during the epidemic have increased significantly compared with before the epidemic, reflecting the increased dependence of enterprises on the internal capital market.

The study will be conducted by: Multi-dimensional visualization to characterize the overall trend of RPT. Quantitative models were used to verify the statistical significance of the increase in trading volume and structural changes. Based on the analysis of industry and regional heterogeneity, identify which companies rely more on internal transaction mechanisms.

1.3 Research Method: Combination of Visualization and Econometric Model

This study breaks away from the traditional regression-only RPT research framework and adopts a "two-stage" approach:

Step1 Visual analysis: Identify structural trends and patterns from the spatial-industry-temporal dimensions.

Step2 Quantitative verification: Ensure the reliability of causal inference through fixed effect and robustness tests.

This approach reinforces the combination of observational intuition and statistical validation, providing systematic evidence for understanding the governance implications of RPT in the context of the pandemic.

In general, the existing literature mainly focuses on the tension between the efficiency-promoting effect and the hollowing out effect of related party transactions. However, systematic quantitative analysis for specific regions (Guangdong) and specific periods (during the epidemic) is still lacking. This study helps to clarify whether enterprises use the internal capital market as a risk buffer mechanism or a channel for resource plunder and benefit transfer under extreme stress situations such as the epidemic, to deepen the understanding of the governance and internal market operation mechanism of Chinese listed companies.

2 Dataset

2.1 Data Source

Primary Databases and Sample Construction

The empirical analysis is based on a multi-dimensional dataset constructed primarily from the CSMAR (China Stock Market & Accounting Research) database. Three core CSMAR tables are employed:

2.1.1 Firm Basic Information Table

This table provides foundational characteristics of listed firms, including industry classification, registered address, establishment date, and other static attributes.

2.1.2 Related Party Transaction (RPT) Master Table

This data set records transaction-level observations of RPTs, documenting transaction type, direction, monetary value, partners, and transaction dates.

2.1.3 Related Party Information Table

This table contains detailed background information on related parties, including their registered addresses and province-level geographic identifiers, enabling spatial matching of RPT flows.

2.2 Data Description

2.2.1 Temporal Coverage

The sample spans **2017–2024**, covering three analytically distinct periods: pre-COVID-19, during COVID-19, and post-COVID-19. This yields a balanced eight-year panel suitable for examining the evolution of RPT patterns across structural shocks.

2.2.2 Sample Scope

The core sample includes all disclosed RPTs of firms listed on the Chinese A-share market, encompassing companies listed on the Main Board, ChiNext, and other major segments.

2.2.3 Integrated Multi-Component Dataset

To support the empirical analysis and to test competing hypotheses regarding “support/insurance” versus “tunneling,” I construct a consolidated dataset comprising three main components.

For more details about the dataset, please refer to the data dictionary below.

Table 2.1 Dictionary of Guangdong Province related party transaction analysis dataset.

Guangdong Province related party transaction analysis dataset		
Column	**Data Type**	**Description**
Stkcd_01	Int64	Main company stock code

Year	Int64	Operating year
Coname_cn_01	object	Name of the main company
Prvn_01	object	The province where the main company is located
Pftn_01	object	The city where the main company is located
Indsub_01	object	Industry information of the main company
Indcodeb_01	object	The industry code of the main company
Isam	Float64	Related party transaction amount
pannrsm	Float64	Proportion of related party transactions
Repat	Int64	Transaction type
Relation	Int64	Relationship
Trasub	Int64	Nature of the transaction
Direction	Int64	Trading direction
Coname_cn_02	object	Name of the related party company
Prvn_02	object	The province where the related party is located
Pftn_02	object	The city where the related party is located
period	object	Classification of epidemic stages

3 Data Processing

The data-cleaning procedure is implemented in multiple stages to ensure consistency, completeness, and analytical reliability across all datasets. The workflow is summarized as follows.

3.1 Data Cleaning

The data processing process is phased in to ensure strict standards for consistency, integrity, and analytical reliability across multiple data sources.

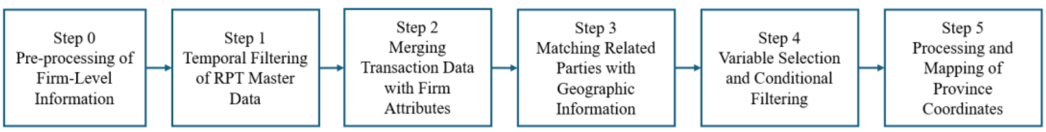

Figure 3.1 Data cleaning process

First, the basic information of the enterprise is preprocessed, including variable standardization, unification of stock code format and date format conversion, and samples that have been delisted or missing key information before 2017 are removed. Second, all records from 2017 and beyond are filtered from the related party transaction master table to align the time window with the study framework. On this basis, the related party transaction record is left connected with the enterprise attribute data according to the standard stock code, so as to introduce the enterprise-level information into the transaction-level data.

The third step is to generate city-level geographical indications by matching the names of related parties and their address information. Due to the high accuracy of geographic information required by spatial analysis, transaction records that could not be matched to valid addresses were eliminated. Subsequently, key variables such as year, industry, city, transaction type and transaction amount were screened according to the research needs, and the samples were further filtered to ensure the comparability of each set of data. Finally, the provincial data are mapped to standardized geographical coordinates to support subsequent spatial visualization and geographical relationship analysis.

In general, this study realizes the complete transformation from original disclosure information to structured analysis data through systematic data integration and hierarchical cleaning, laying a solid data foundation for subsequent empirical research.

3.2 Variable Construction

Following data cleaning, a series of derived variables and analytical indicators are constructed to support the visualization, statistical modeling, and mechanism-testing components of the study. Feature engineering focuses on three major dimensions: **temporal**, **industry/transactional**, and **spatial/network** attributes.

3.2.1 Temporal Variables

To align with the study’s three-stage design, we generate standardized time indicators:

1) Year: extracted from the reporting date (Reptdt), serving as the primary temporal index.

2) Period Classification:

a. *Pre- COVID*: 2017–2019

b. *During-COVID*: 2020–2022

c. *Post-COVID*: 2023–2024

This categorical variable enables segmented visualizations and difference-in-differences style comparisons of RPT behavior across crisis phases.

3.2.2 Spatial and Network Features

1) Spatial attributes enable the construction of geographic trade flows and region-level dependency measures, including Province Coordinates and Regional Categorization.

2) Network indicators quantify the structural positions of firms within RPT networks: Degree Centrality, betweenness centrality and Weighted Degree.

These features are essential for identifying systemic risk propagation paths, group-level dependencies, and spatial clustering.

4 Visualization Design

In this study, Plotly and Dash jointly implement an interactive web app to systematically present the spatio-temporal evolution characteristics of related party transactions from 2017 to 2024 by constructing a multi-module and interactive visual analysis board. The overall design follows a hierarchical philosophy of "big overview - subdivision - drill into details", allowing users to grasp trends at the macro level and analyze structural characteristics at the industry, enterprise and regional levels when needed.

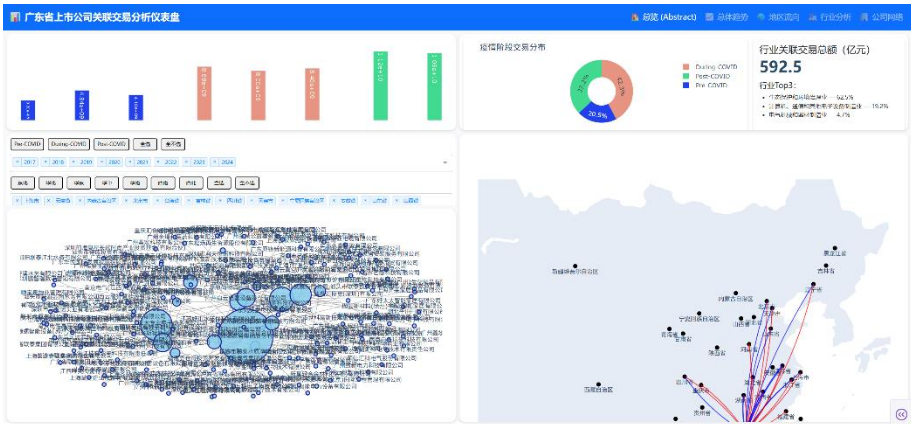

Figure 4.1 System Overview

In the Abstract summary diagram displayed on the default homepage, the key ideas of the four core modules are integrated, including:

a) Overview – Trade Value Time Variation

b) Company Network - Related-Party Transaction Network

c) Industry View - Total Amount of Industry Transactions

d) Region Flow – Interregional Trade Flow Map

The central area of the panel board is set up with interactive filtering tools for "Year" and "Region", and users can independently select specific years and provinces through the "Select/Deselect" operation, so that the four modules can be refreshed in conjunction to support comparative analysis.

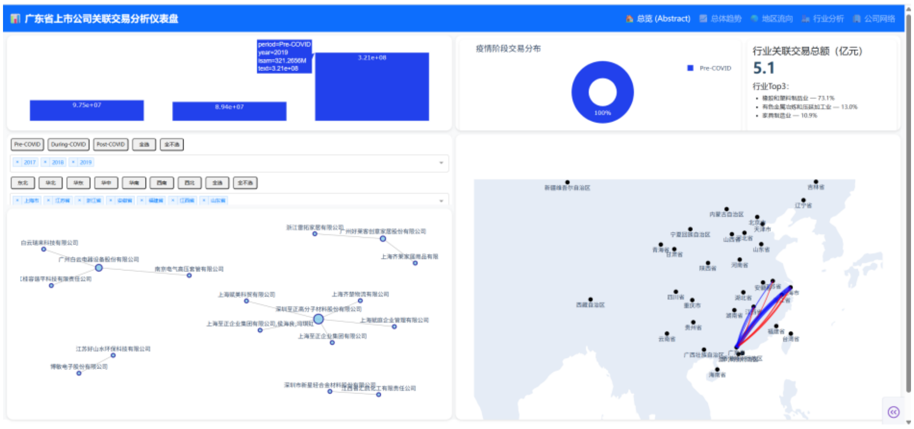

Figure 4.2 System Overview after Select Interaction

4.1 Overview: Trade Value Time Variation

Design:

The annual transaction size is visualized in combination with bar charts and line charts. The bar chart represents the total amount of related party transactions in each year, and the line chart depicts the trend change and shows the time series structure from 2017 to 2024. (1) The horizontal axis is the year (2017-2024) (2) The vertical axis is the annual transaction amount (3) Different colors indicate the three stages before, during and after the epidemic to enhance stage identification.

Figure 4.3 Trade Value Change with Time Variation

Interaction:

When a user selects a specific year, the data in the other modules will be updated synchronously. When the user's mouse hovers over the histogram/line chart, the specific transaction data is displayed.

Figure 4.4 Trade Value Change with Time Variation after Selecting

4.2 Company Network: Related-Party Transaction Network

Design：

The related party transaction network between enterprises is presented in a force-oriented layout: (1) the size of nodes (circles) represents the total transaction size of the enterprise; (2) Lines represent the transaction relationship between enterprises, and the existence and thickness of edges reflect the existence and strength of transactions; (3) The spatial layout between nodes can reveal the internal structure of the enterprise group, key hub enterprises and potential systemic risk points.

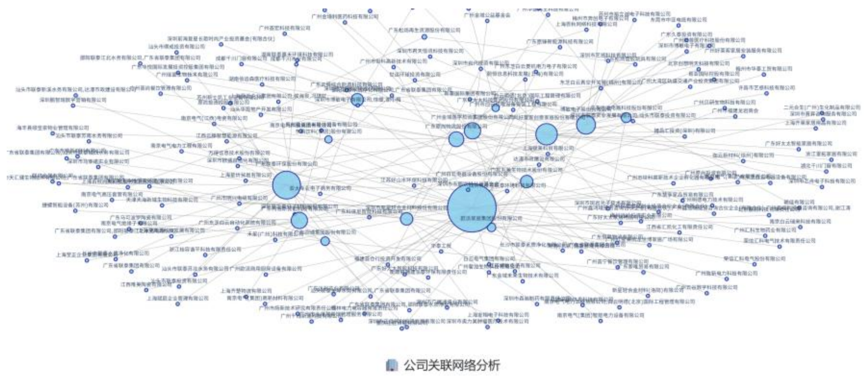

Figure 4.5 Related-Party Transaction Network

Interaction:

Right-click and drag to zoom in on the local area; Double-click the right button to revert to the default view; As the user selects a year or region, the network structure is automatically filtered and rearranged.

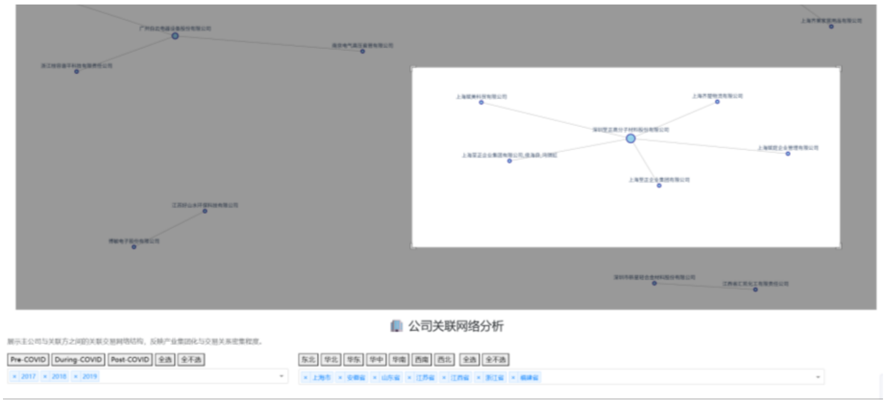

Figure 4.6 Related-Party Transaction Network with zooming interaction

4.3 Industry View: Total amount of industry transactions

Design:

The pie chart is used to show the relative scale of related party transactions of each industry and distinguish different stages (before/during/after the epidemic) by color, clearly showing the structural differences between industries. The chart on the right above shows the top 3 industries and their proportions, highlighting the main contributing industries and helping to identify structural changes, such as whether the proportion of transactions in manufacturing or environmental governance industries has increased significantly during the pandemic.

The middle part shows the top ten industries in terms of related party transaction amount, and the stack chart shows the total transaction amount accumulated during a specified year, distinguishing different epidemic stages by color.

Figure 4.7 Total amount of industry transactions

Interaction:

The industry pie chart is synchronized with the year filter and the total volume chart, and the industry share structure will be updated immediately when the user switches years.

Figure 4.8 Total amount of industry transactions

4.4 Region flow: Interregional Trade Flow Map

Design:

Cross-regional transactions are presented as Flow Maps: (1) the city where the listed company is located in Guangdong Province is used as the starting point of the flow direction; (2) The location of the affiliated company is the end point; (3) The thickness of the line indicates the size of the transaction amount; (4) Color distinguishes the trading direction (capital inflow or outflow); (5) The geospatial layout shows the strength and weakness relationship of related party transactions between regions, and intuitively reflects regional dependence, core nodes and potential spatial risk concentration. (6) The heat map provides a numerical comparison between different provinces.

The diagram transforms abstract transaction relationships into spatialized visual expressions, which helps to understand the capital connection network between industries and regions and their dynamic evolution with the epidemic.

Figure 4.9 Interregional Trade Flow Map

Interaction:

After the user selects a specific year or region, the flow line in the map will automatically update to form a dynamic spatial visualization that can be filtered, zoomed in, and compared. When the mouse hovers over the line, the direction of the transaction between provinces and the transaction amount can be observed.

Figure 4.10 Interregional Trade Flow Map after Selecting Interaction

5 Evaluation

5.1 Temporal Patterns in Related-Party Transactions

According to the visual comparison of the line chart on the visual panel, the trend of the total annual related party transaction shows the growth of the total amount at different stages.

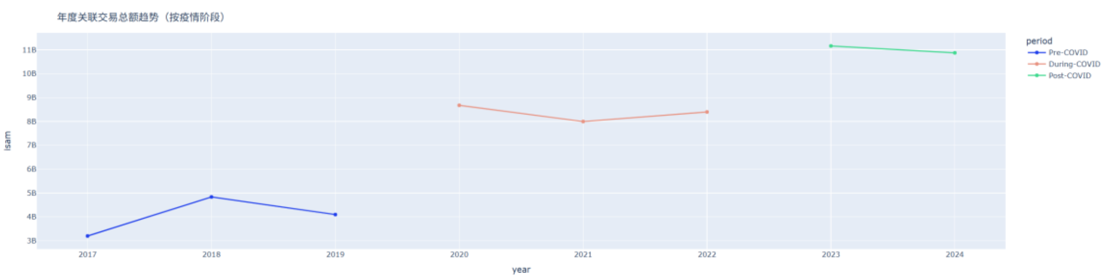

Figure 5.1 The Trend of Total Annual Related Party Transactions at Different Stages of the Epidemic

To evaluate whether the impact of the epidemic has changed the related party transaction behavior of listed companies in Guangdong Province, a regression model is constructed based on transaction value (ISAM) to determine whether related party transactions are more in line with the "tunneling" or "buffering" mechanism of risk sharing.

The core regression model is set as follows:

$$isam_{it} = \beta_1 During + \beta_2 Post + \beta_3 relation + \beta_4 repat + FE_{industry} + \epsilon$$

Among them, During and Post identify observations during and after the pandemic, respectively, Relation represents the type of association, and Repat represents the category of transactions. If the During coefficient is significantly positive during the epidemic, it means that the enterprise may increase upstream or internal financial support through related party transactions, thereby reflecting the risk-sharing function. If the During is positive and is primarily concentrated in the parent company or major shareholder relationship, it may reflect the flow of resources to the controlling party, which is in line with the tunnel effect logic. In addition, if the size of the relationship (e.g., the proportion of PANRNRSM) increases significantly with the pandemic, it indicates an increase in dependence on related parties.

(1) Baseline regression results: The epidemic did not significantly change the amount of RPT. The benchmark regression results showed that the dummy variables during and after the epidemic failed the significance test. Specifically, the During coefficient is 7.63×10^6 ($p = 0.601$) and the Post coefficient is 1.57×10^7 ($p = 0.264$). This result shows that the COVID-19 outbreak has not significantly increased or reduced the amount of related party transactions. Guangdong listed companies have neither significantly expanded internal transactions to reallocate resources during the epidemic, nor have they reduced the scale of related party transactions due to governance pressure or tightening regulations.

Relation and transaction category also did not have significant impact. The p-value of Relation is 0.206 and the Repat is 0.390, indicating that the amount of related party transactions is mainly driven by the normal operating needs of the enterprise, rather than determined by the control structure or special transaction types. Overall, the impact of the epidemic has not systematically changed the related party transaction behavior of enterprises, nor has it triggered a significant tunnel effect or internal capital support effect.

(2) Industry differences are the most significant explanatory factors for related party transactions. Among the fixed effects of all industries, most industries have a significant positive impact compared with the benchmark industry, especially ecological protection and environmental governance, electronic equipment manufacturing, non-ferrous metal smelting and processing industry, etc. These industries are generally more capital-intensive and have more complex supply chain structures, so their RPT amounts are significantly higher than other industries.

According to the value of the total amount of related party transactions in the panel, the top 3 industries account for 86.4% of all industries, which is a high value.

Figure 5.2 Transaction amount Industry Distribution Figure

This finding shows that the key mechanism to determine the scale of related party transactions is not the impact of the epidemic, but the characteristics of the industry and the structure of the supply chain. The related party transaction model of listed companies in Guangdong Province is closer to the dependence of the industrial chain and the law of industry operation, rather than changes in governance strategies or profit transfer machines.

(3) Robustness test: To enhance the reliability of benchmark conclusions, several robustness tests were carried out, such as logarithmic processing, tailing processing, and correlation group regression. After logarithmic processing, the results are more stable, and the explanatory power is improved. The result is consistent after the tailing treatment excludes the influence of the outliers. Group regression based on correlation relationships shows that neither the parent company nor the non-parent company has been affected by the epidemic.

(4) Event Study analysis: There is no short-term or long-term impact of the epidemic. An event study model of OLS lead/lag was used to test from three years before and after the event (t = –3 to +3). The results showed that the pre-event trend was stable, with no significant jump in the year of the epidemic and no continuous change 1–3 years after the event. Overall, the impact of the epidemic on the amount of related party transactions was close to zero before, at and after the event.

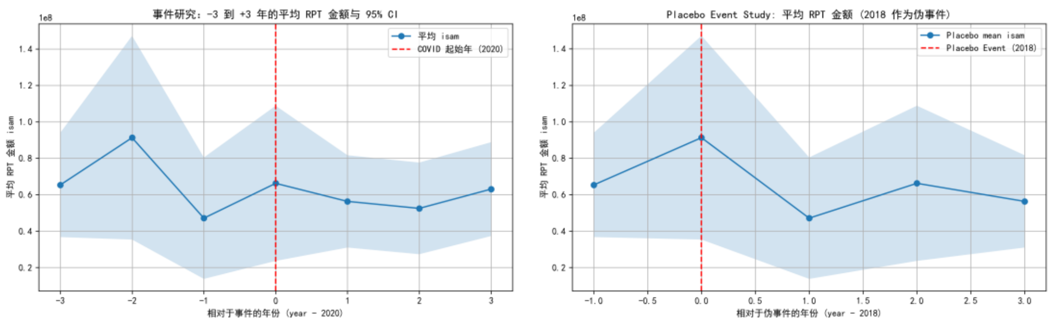

Figure 5.3 Event Study Analysis Figure

(5) Placebo test: There is no abnormal beating of pseudo-events. Re-estimating 2018 as a pseudo-event yields a nearly identical volatility structure with no significant jumps. The results show that the fluctuations in the data are due to their own characteristics, not the impact of the epidemic, which further supports the above conclusion.

Based on benchmark regression, robustness test, event research and Placebo test, the overall judgment of the relationship between the epidemic and related party transactions is highly consistent. This study found no evidence of tunnel effect or internal capital transmission, and the related party transaction behavior of enterprises in Guangdong Province is more in line with the logic of industrial chain demand and supply chain structure, rather than abnormal governance behavior induced by the epidemic.

5.2 Spatial Patterns in Related-Party Transactions

Based on the company's network dashboard, compared with three different periods, the amount of related party transactions of the company under the impact of the epidemic is highly condensed to several specific companies, reflecting the state dominated by core enterprises. However, in the post-epidemic stage, there was a structural decline.

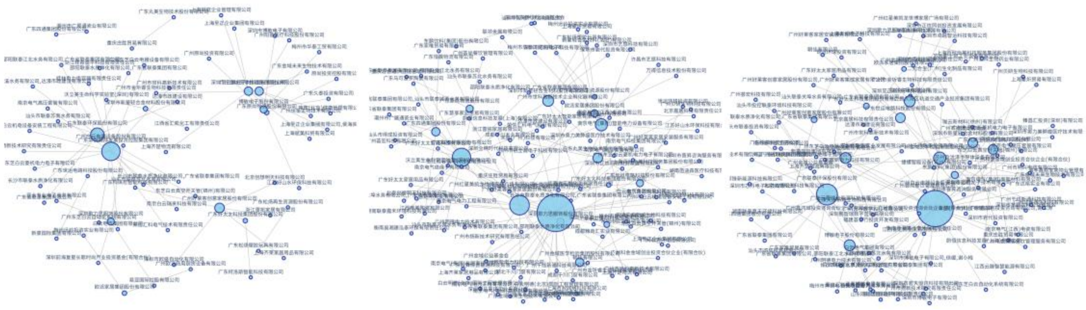

Figure 5.4 Comparison of Company Networks at Different Stages of the Epidemic

Based on the company-supplier dichotomous network and its company projection network based on the Related-Party Transactions (RPT) data of listed companies in Guangdong Province from 2017 to 2024, this study systematically analyzes the annual centrality (strength, eigenvector), network concentration (Top 10 proportion, Gini coefficient) and annual network topology. The overall results show a highly consistent structural pattern that reflects the formation mechanism of the RPT network and its dynamic evolution path during the sample period. The main findings are as follows:

5.2.1 The overall network scale shows a cyclical structure of "generation-expansion-peak-contraction"

A typical structural cycle can be observed from the annual total strength change: the network began to form in 2018, peaked in 2020–2021, weakened significantly after 2023, and almost degenerated into inactivity by 2024.

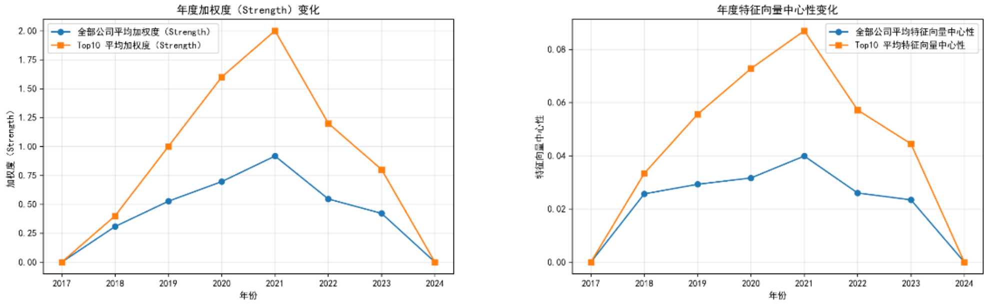

Figure 5.5 Line Chart of Change in Annual Strength and Eigenvector Centrality

This trend reflects a significant sensitivity of RPT behavior to the external environment. During the epidemic, supply chain uncertainty has increased significantly, and enterprises tend to maintain operational stability through "internalized transactions" and "intra-group collaboration", thereby promoting the intensification and scale expansion of related party transaction networks. After the impact of the epidemic eased, the intensity of RPT gradually declined, indicating that the previous expansion was more of a "stress-type structure" and gradually returned to normal after the external shock disappeared.

5.2.2 Network concentration has been at a very high level for a long time, showing a typical core-edge structure

The proportion of Top 10 nodes remained between 90% and 100% in most years, indicating that the RPT network is highly concentrated in a small number of enterprises. This structure means: (1) the transaction volume of the network is almost completely controlled by the leading enterprises; (2) the participation of most listed companies in the RPT network is extremely low; (3) The network connection mode is not decentralized but presents obvious structural characteristics of "center-dominated and peripheral sparse".

This pattern belongs to the typical core-periphery topology in economic networks, indicating that RPT behavior is manifested as an unbalanced structure of highly concentrated leading enterprises in Guangdong Province, rather than a universal and wide-coverage related party transaction system.

The central structure at the node level is highly stable, and Guangzhou Baiyun Electric Equipment Co., Ltd. is a long-term core enterprise. As can be seen from the annual trajectory of strength and eigenvector centrality, Guangzhou Baiyun Electric Equipment Co., Ltd. has been at the absolute center of the network in almost all years. This means that the company not only has a large scale of related party transactions but also is in the structural position of "connecting high-impact nodes” and is a leading enterprise that maintains the stability and connectivity of the entire RPT network.

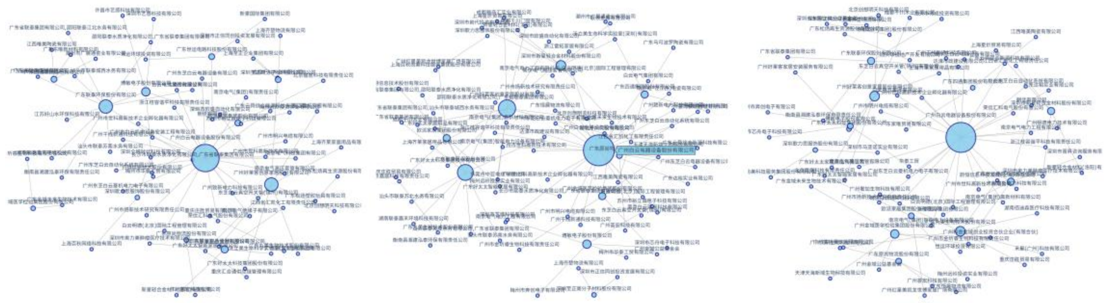

Figure 5.6 Comparison Chart of the Company's Network for 3 Years (2020, 2021, 2022) During the Epidemic

2021 was the strongest year for the network, and the structural scale, concentration, and centrality reached historical peaks. This structural peak highly coincides with the strongest period of the epidemic's impact on the supply chain, showing a significant "agglomeration effect" of enterprises responding to external risks through related party transactions.

5.2.3 Comprehensive judgment: core enterprises leading, agglomeration during the epidemic, and structural decline in the post-epidemic period

On the whole, the RPT network of listed companies in Guangdong Province shows the following pattern: The evolution of RPT network shows the typical characteristics of "crisis-driven agglomeration": it is highly cohesive under the impact of the epidemic, and the structure gradually returns to normal in the post-epidemic period, but the dominance of core enterprises has not been completely weakened, indicating that the control of the supply chain is inertia and path dependence.

5.3 Case Study: Analysis of Heterogeneity at the Industry and Provincial Level

This section examines the impact of the core explanatory variable PANNRSM on ISAM from the perspectives of industry and province, to identify the differences in related party transaction behavior in different industry structures and regional institutional environments. Based on the group regression results, the analysis focused on the evidence in the direction of regression coefficient, significance, annual trend and sample robustness.

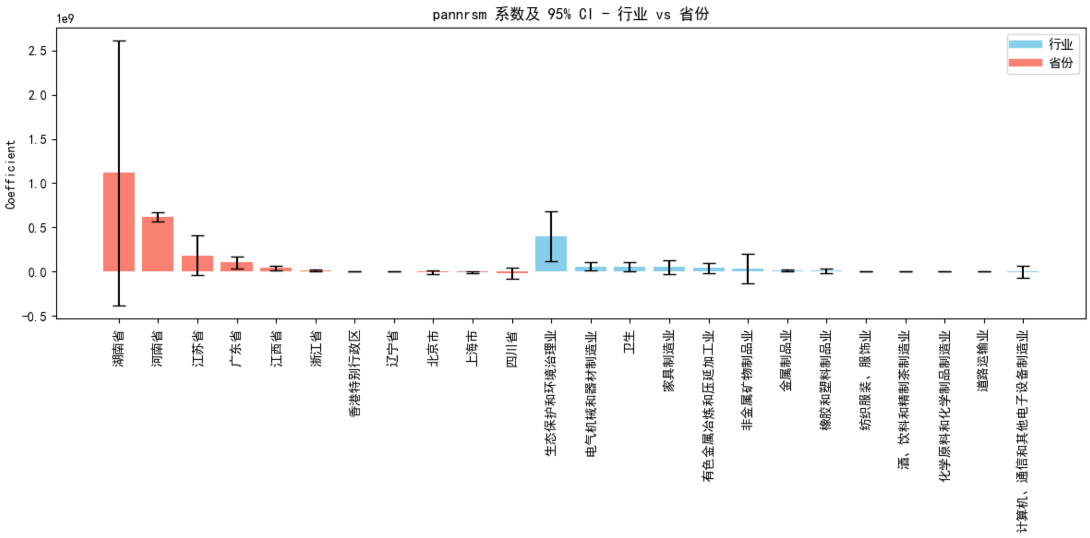

Figure 5.7 Comparison Chart of PANNRSM Coefficients between Industries and Provinces

5.3.1 Industry Perspective: The Structural Heterogeneity of the Impact of Related Party Transactions

(1) The impact of PANNRSM on the industry difference in related party transaction amounts

In the industry group regression, the regression coefficient and significance of PANNRSM showed significant cross-industry differences. First, many industries show a significant positive relationship, including chemical raw materials and chemical products manufacturing, health industry, ecological environment management, electrical machinery manufacturing, textile and garment industry, wine and beverage manufacturing industry, and metal products industry. The results show that in these industries, the increase in the proportion of related party transactions will significantly promote the increase in the amount of related party transactions. This often reflects strong supply chain dependencies within the industry, the need for collaboration within the group, or high inertia in internal transactions.

Secondly, the estimated coefficients of some industries are positive but not significant, such as furniture manufacturing, non-ferrous metal processing, rubber and plastic products, and non-metallic mineral products. This means that PANRNRSM has limited explanation of transaction amounts in these industries, and its effectiveness may be weakened by differences in industry size, corporate governance structure, or business model.

In addition, a few industries (such as computer, communication and electronic equipment manufacturing) showed a small negative coefficient. This type of industry usually has a high degree of marketization and competitive pressure, and the impact of related party transactions on the overall transaction scale is unstable, which may be affected by individual strategies, industry cyclical or technological shocks.

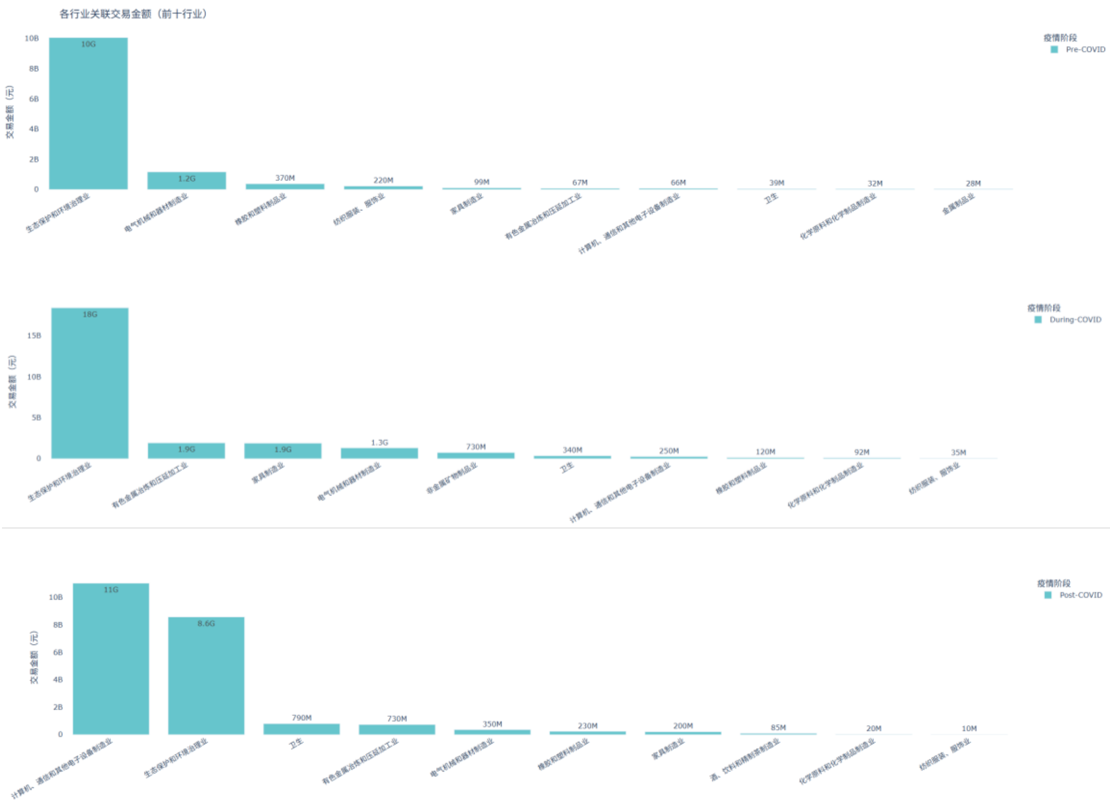

Figure 5.8 Industry Transaction Amounts at Different Stages of the Epidemic

Overall, the sensitivity of PANRSM varies greatly among industries, for example, the coefficient of the ecological and environmental governance industry reaches about 400 million yuan, while the road transport industry is only about 700,000 yuan. This indicates that there are significant differences in the mechanism of related party transactions in different industrial structures.

(2) Year fixed effect: the time series structure of the industry transaction amount

The fixed effect of the year reveals the systemic impact of the epidemic and policy cycle on related party transaction behavior. The main trends are as follows:

Table 5.1 The Systemic Impact of the Epidemic on Industries

Industry	The Systemic Impact of the Epidemic
Chemical raw materials and chemical products industry	From 2021 to 2024, there was a downward trend, and related party transaction activity contracted significantly in the post-epidemic period.
Health industry	The coefficient continued to rise and was significant in each year, reflecting the expansion of the scale of

	related party transactions due to the increase in demand in the medical industry during the epidemic and post-epidemic stages.
Textile and garment industry	The year coefficient is significantly negative and highly significant, indicating that the industry experienced chain contraction or a decline in demand during the sample period.
Electronic information industry	The year's effect is volatile, indicating that the industry is more susceptible to technological updates or market events.
Ecological and environmental governance industry	The number of related party transactions increased significantly in 2020-2021 and declined sharply in 2023, indicating that the industry is facing policy-driven fluctuations.

In summary, the annual trend shows distinct industry heterogeneity, reflecting the differences in the organizational structure of the industrial chain and the behavior model of enterprises, and reflecting the different sensitivity of the industry in the structural shock of the epidemic, policy orientation or market cycle. At the same time, there is a significant corporate fixation effect in some industries, indicating that the internal governance structure of enterprises has a key impact on related party transaction behavior.

5.3.2 Provincial Perspective: Regional Institutional Differences in Related Party Transaction Behavior

The inter-provincial regression results showed that there was significant regional heterogeneity in the impact of PANNRSM, indicating that the institutional environment, regulatory intensity, industrial structure and distribution of group enterprises in different regions played an important role.

Figure 5.9 Inter-regional Transaction Flow at Different Stages of the Epidemic

(1) Inter-provincial heterogeneity: influence direction and significance grouping

Based on the provincial regression coefficient, the regional characteristics can be summarized as follows:

Provinces with significant positive impact: Guangdong, Jiangxi, Zhejiang, Henan, and Hong Kong Special Administrative Region.

These provinces have highly consistent structural characteristics: manufacturing and industrial chains are highly concentrated; There are many group enterprises, and the internal cooperation mechanism is obvious; Related party transactions may assume supply chain stability or intra-group regulation functions. Among them, Henan's coefficient is extremely large, which may reflect significant transmission or profit adjustment behavior.

The proportion of related party transactions has a strong explanatory power on the transaction amount, reflecting the high concentration of the industrial chain, group collaboration or tunnel behavior is more obvious.

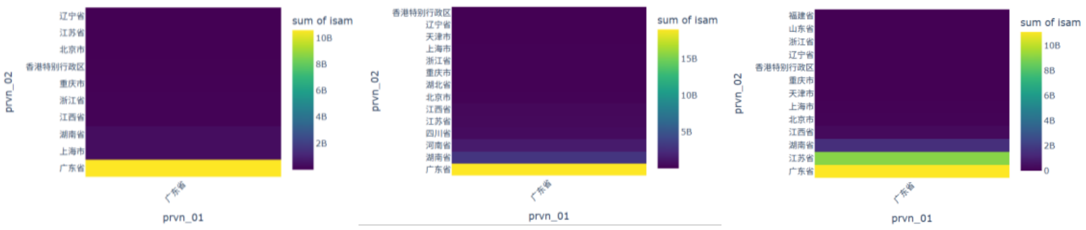

Figure 5.10 Heat Map of Transaction Amount Between the Main Company and Related Party Provinces

Insignificant provinces: Shanghai, Beijing, Sichuan, Jiangsu, Hunan, Liaoning.

This group of provinces presents opposite structural characteristics, including: a high proportion of state-owned enterprises and stricter supervision; The industrial structure is diversified and related party transactions are restricted; the degree of marketization is high, and the transparency of internal transactions within enterprises is higher; The sample size of some provinces is small, which affects the significant performance. The structural characteristics of this group indicate that in regions with strong regulatory strength or more mature corporate governance, the proportion of related party transactions does not necessarily lead to an increase in transaction amount.

(2) Mechanism explanation of inter-provincial differences

Significantly positive provinces reflect two types of potential mechanisms:

The first category: scale effect and internal collaboration motivation. The supply chain relationship in the provinces where manufacturing is concentrated is close, and listed companies rely on internal networks to stabilize production, and related party transactions have a functional role.

The second category: potential tunneling behavior risks. The coefficient in some provinces, such as Henan, may reflect the phenomenon of profit adjustment or related transmission.

Insignificant or negative provinces may reflect: a strong regulatory environment constraints related party transaction; High transparency leads to more standardized related party transactions; the industrial structure is scattered, and the dependence on internal transactions is weak; sample size limits statistical significance.

In general, the related party transaction behavior between provinces shows significant structural differences. Related party transactions are not only the result of corporate behavior, but also profoundly affected by factors such as regional industrial structure, regulatory intensity, and government governance model.

6 Conclusion

Based on the long-panel data of related party transactions (RPTs) of listed companies in Guangdong Province from 2017 to 2024, combined with the integrated framework of network analysis, industry grouping, regional characteristic modeling, and multi-module visual dashboards, this study systematically reveals the evolution logic and heterogeneity of RPT in Guangdong Province from the temporal series, structure, industry and spatial dimensions. The main conclusions are as follows.

Firstly, from the perspective of the overall network structure and time series trends, the long-panel data enables this study to accurately identify the dynamic change patterns before, during and after the epidemic under the premise of strictly controlling the annual fixed effect. The results show that the related party transaction network exhibits a typical core-edge structure and expanded rapidly during the epidemic period from 2020 to 2021, reaching the peak of network strength and transaction amount, and then falling back during the weakening period of the epidemic, forming a cyclical evolution characteristic of "epidemic stimulus expansion - post-epidemic contraction". Through the Overview: Trade Value Time Variation module in the visual dashboard, combined with the timing presentation of the bar combination chart, the trend of network contraction and expansion can be intuitively seen. The concentration and inequality of the RPT network have remained at a very high level for a long time, with the proportion of transactions by the top 10 nodes approaching 100%, and the Gini coefficient stabilizing in the range of 0.70-0.80. The network visualization module further demonstrates the significant centrality of Guangzhou Baiyun Electric Co., Ltd.

as a core node and verifies that the core enterprises still have structural inertia and continuous dominance under the impact of disturbances.

Secondly, industry panel regression and visual analysis jointly reveal significant industrial structure heterogeneity. The long-panel model makes the estimation results more explanatory and robust by controlling annual changes, industry characteristics and enterprise size differences. The empirical results show that in industries such as chemical products, health, electrical machinery, ecological and environmental protection, textile and clothing, beverage manufacturing, etc., the proportion of related party transactions (PANNRSM) has a significant positive impact on the transaction amount (ISAM), indicating that industries with high dependence on the industrial chain or strong internal synergy mechanism are more likely to form the expansion effect of related party transactions. On the contrary, industries such as computer electronics, rubber and plastics, and non-metallic products are weakly correlated or insignificant, reflecting that in industries with a high degree of marketization and strong external substitution, the impact of related party transactions on business activities is more limited. The Industry View module in the visual dashboard displays the differences in industry contribution and the changes in the industry structure under the impact of the epidemic through pie charts and Top 3 industries, making it easier to observe and compare the differences in industry contribution and the changes in the industry structure under the impact of the epidemic.

Thirdly, the inter-provincial panel analysis reveals obvious regional institutional differences. Relying on the cross-section-time joint structure of the 8-year period, the regional fixed effect can effectively eliminate the characteristics of regional cohabitation, thereby highlighting the impact of the institutional environment, regulatory strength, and industrial base on related party transactions. The study found that in regions with high manufacturing concentration or high density of private groups, such as Guangdong, Jiangxi, Zhejiang, Henan, and Hong Kong, PANNRSM significantly pushed up ISAM, indicating that related party transactions have scale effects and synergies in these regions. In provinces with stricter regulatory environments or high concentrations of state-owned enterprises, such as Beijing, Shanghai, Jiangsu, and Sichuan, the relationship between PANNRSM and ISAM is not significant, indicating that strong supervision and diversified industrial structure can significantly restrain internal transaction dependence. The Region Flow map of the visual Kanban presents the flow of funds and transactions across provinces, so that the structural transaction pattern generated by regional institutional differences is concretely presented in the form of a spatial network.

In general, this study integrates the quantitative recognition ability of long-panel data with the cognitive enhancement ability of multi-module visual kanban, so that the structural, dynamic and spatial characteristics of related party transactions can be presented in multiple dimensions. The results show that related party transactions are not homogeneous economic behaviors, but are affected by the interaction of enterprise scale, industrial chain location, regional institutional environment and epidemic shock. During the epidemic, related party transactions have been strengthened in stages, but the network advantages of core enterprises have significant path dependence, and the differences from industry to region further highlight the complex institutional and industrial logic in China's enterprise group system.

This study has important implications for understanding the governance mechanism, supply chain resilience and network organizational structure of related party transactions of listed companies in China. Future research can combine more fine-grained corporate governance data, policy impact experimental design, equity network and group hierarchy to explore the role of related party transactions in risk transmission, resource allocation and corporate performance from the mechanism level, so as to further deepen the systematic understanding of China's enterprise network economy.

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